MINI PROJECT REPORT

ADVANCE MICROCONTROLLER

PROJECT TITLE

WiFi Controlled Car usign ESP8266

PROJECT GROUP DETAILS

Sr. No.	Enrollment No.	Student Name
1	250283111021	Soni Krish
2	250283111022	Yug Shah
3	250283111016	Patel Heet Rohitkumar

PROJECT DESCRIPTION & PROTOTYPE

1. PROJECT WORKING

The project uses an ESP8266 NodeMCU microcontroller to control a car wirelessly through a Wi-Fi-connected Android application called Smart Car Controller. Commands sent from the mobile application are received by the ESP8266 and processed to control the motor driver. The motor driver operates two DC motors to move the car forward, backward, left, and right. The speed can be adjusted using PWM (Pulse Width Modulation), and a buzzer is used as a horn. Thus, the car can be controlled remotely using a smartphone over Wi-Fi.

2. APPLICATIONS

1. Wireless robotic car control and educational demonstrations.
2. Remote-controlled vehicles for basic surveillance and inspection.
3. Learning embedded systems, Wi-Fi communication, and motor control.

3. MAJOR COMPONENTS

Microcontroller: NODE-MCU-ESP-8266

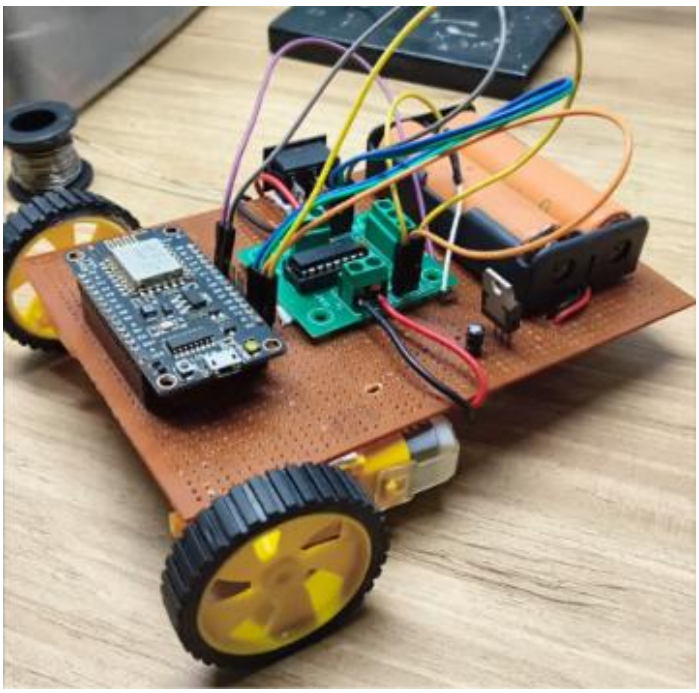
Sensors: N/A

Actuator / Display: DC Motors, Mobile Controller

Other major component: L293D IC,

Software: Arduino IDE, MIT App Inverter

4. PROTOTYPE / WORKING MODEL PHOTOGRAPH



5. PROJECT/ PROTOTYPE NAME

Wi-Fi Controlled Smart Car Using ESP8266

PROJECT OUTCOME / EVIDENCE

KEY OUTPUT / RESULT

Successfully developed a Wi-Fi-controlled car using ESP8266 NodeMCU. The car can move forward, backward, left, and right using an Android application. Speed control is implemented using PWM, and a buzzer provides a horn function.

Mapped SDG (Sustainable Development Goal)

SDG 9 – Industry, Innovation and Infrastructure